



APRO National Safety Code (NSOC) – v1.0

Critical safety protocols for all launch operations. Mandatory reading for all members, engineers, and participating student societies.

1. Regulatory Compliance & General Principles

Prime Responsibility: The ultimate responsibility for safety rests with the organization or individuals operating the facilities and activities. Adequate financial and human resources must be allocated to ensure a strong safety culture.

PSARB Authorization: No space activity, including launches, may commence within the national territory without explicit authorization and registration from the Pakistan Space Activities Regulatory Board (PSARB).

Public Safety: Launch and testing activities must never pose an immediate threat to public order, public health, or the safety of persons and property.

Environmental Protection: Operations must avoid any adverse environmental changes or harmful contamination.

Optimization of Protection: Risks must be assessed throughout the lifetime of the launch facility or activity using a graded approach to avoid hazards as far as practicable.

2. Pre-Launch & Site Operations

Airspace Clearance: All launch operations require strict CAA NOTAM (Notice to Airmen) management prior to ignition to ensure clear airspace. * **Launch Pad Infrastructure:** Launch pads must be structurally validated for the specific rocket class, such as accommodating a maximum length of 3 meters for standard high-power model rockets.

Security Vetting: Prior to collaborating with or procuring services from any foreign organization or firm for space-related activities, clearance must be obtained from concerned security agencies through the registration cell.

3. Propulsion & Static Testing Safety

Standoff Distances: A minimum safety standoff distance of 50 meters must be strictly maintained by all personnel during static motor testing. * **Ignition Systems:** All motor ignitions, both on the test stand and the launch pad, must utilize a remote relay ignition system.

Deflection & Shielding: Static test stands must be securely mounted to a concrete pad using a welded steel frame and must incorporate a steel blast deflector.

Motor Casing Margins: Aluminum 6061-T6 motor casings (e.g., 75mm OD assemblies) must maintain a safety factor greater than 3×.

4. Flight, Recovery & Termination

Recovery Deployment: High-powered rockets must utilize a verified dual deployment recovery system to ensure controlled descent, specifically deploying the drogue parachute at apogee and the main parachute at a predetermined safe altitude (e.g., 300 meters).

Redundant Avionics: Flight computers must feature redundancy. Primary systems (e.g., Teensy 4.0) must be backed up by a secondary system (e.g., Arduino Nano) to guarantee deployment charge firing.

Orderly Termination: APRO and its operators are obligated to ensure the orderly and safe termination of all space activities in accordance with local and international guidelines.

5. Liability & Emergency Preparedness

Emergency Response: Necessary infrastructure must be established to coordinate and respond to any facility-level emergencies, effectively mitigating risks.

Insurance & Damages: APRO must secure appropriate insurance to cover potential liabilities for damages caused to persons or property during operations, ensuring affected parties can be compensated. Run-off liability cannot be excluded. *(Note: PSARB may reduce this requirement if the activity is deemed in the public interest for science, research, or education).*